

Lab 6 - 2B

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February 26, 2026

1 Parallel Histogram Computation Using Pthreads

1.1 Background: What Is a Histogram?

A **histogram** is a data structure that counts how many times each possible value appears in a dataset. In this lab:

- The histogram is an array where:

$\text{histogram}[v] = \text{number of times value } v \text{ appears in the dataset}$

1.2 Problem Description

You are given a four-dimensional data array:

`data_array[SUB_ARRAY_COUNT][D][D][D]`

Each sub-array is conceptually 3 dimensional and is processed independently by a separate thread.

- Each thread computes a local histogram for one sub-array.
- Local histograms are merged into a **global histogram**.
- Synchronization must be handled correctly to avoid race conditions.

1.3 Constraints

- **Only one mutex is allowed in the entire program.**

1.4 Provided Code: Data Initialization

The following function initializes the data array with pseudo-random values. This code is provided and must not be modified.

```
void init_data_array() {
    srand(0);
    for (int i = 0; i < SUB_ARRAY_COUNT; i++) {
        for (int j = 0; j < SUB_ARRAY_DIM; j++) {
            for (int k = 0; k < SUB_ARRAY_DIM; k++) {
                for (int l = 0; l < SUB_ARRAY_DIM; l++) {
                    data_array[i][j][k][l] = rand() % RANGE;
                }
            }
        }
    }
}
```

1.5 Tasks

1. Create `SUB_ARRAY_COUNT` threads.
2. Assign each thread exactly one sub-array to process.
3. Compute a local histogram inside each thread.
4. Use the **single mutex** to safely merge the local histogram into the global histogram.
5. After all threads complete, verify that the total histogram count equals:

$$\text{SUB_ARRAY_COUNT} \times D \times D \times D$$

2 H₂O Bond Formation (Barrier)

2.1 Problem statement

A water molecule is formed by 2 Hydrogen atoms and 1 Oxygen atom. Threads represent individual atoms.

2.2 Rules

- H atoms must wait for another H and one O.
- O atoms must wait for two H atoms.
- They must "bond" (exit the barrier) simultaneously in groups of three (H-H-O).
- Hint: You can concurrently create H but not O.

2.3 Constraints

- Use **semaphores only**.
- No mutexes.
- Ensure no atom is left behind or grouped incorrectly.

2.3.1 Expected Execution

H Ready -> H Ready -> O Ready -> Bond Formed!